



ACLC College of Tacloban
Information Technology Department

Final Output In Data Communication And Networking 3

Title: "Designing a Secure Multi-Subnet Network with Justified
Security Techniques"

Submitted by;
Guillena, Jasmine L.
BSIT-3A

Submitted to:
Cardaña, Rens Joshua M.
Instructor

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INTRODUCTION

A computer network allows many devices such as computers, printers, and servers to communicate with each other and share resources like files, data, and internet access. These networks are widely used in schools and offices where many users need to work at the same time. Using a network makes communication faster and more efficient, but it also creates risks if there are no security rules in place. When many users share one network, sensitive information can be exposed if access is not properly controlled.

In schools and offices, users are grouped into different departments such as Human Resources (HR) and Information Technology (IT). Each department has different responsibilities and does not need access to the same information. For example, HR may handle personal records, while IT manages technical systems. Because of this, not all users should be allowed to access every part of the network. Servers usually store important data and services, so they must be protected from unauthorized users. Network security helps control who can access specific resources and prevents misuse of the system.

The purpose of this activity is to design a secure and organized computer network using Cisco Packet Tracer for the subject Data Communication and Networking 3. Cisco Packet Tracer is a simulation tool that allows students to design and test networks without using real hardware. Through this activity, students apply basic networking concepts such as subnetting, routing, and security configuration in a practical and easy-to-understand way. In this network design, Virtual Local Area Networks (VLANs) are used to separate users based on their department. VLANs help reduce unnecessary network traffic and improve organization.

Routing is then used so that devices from different VLANs can communicate only when it is allowed. An Access Control List (ACL) is also implemented to control network traffic by allowing or denying access based on set rules. By combining VLANs, routing, and ACLs, the network becomes more secure, organized, and efficient. This activity helps students understand how simple security techniques can greatly improve the safety and reliability of a computer network.

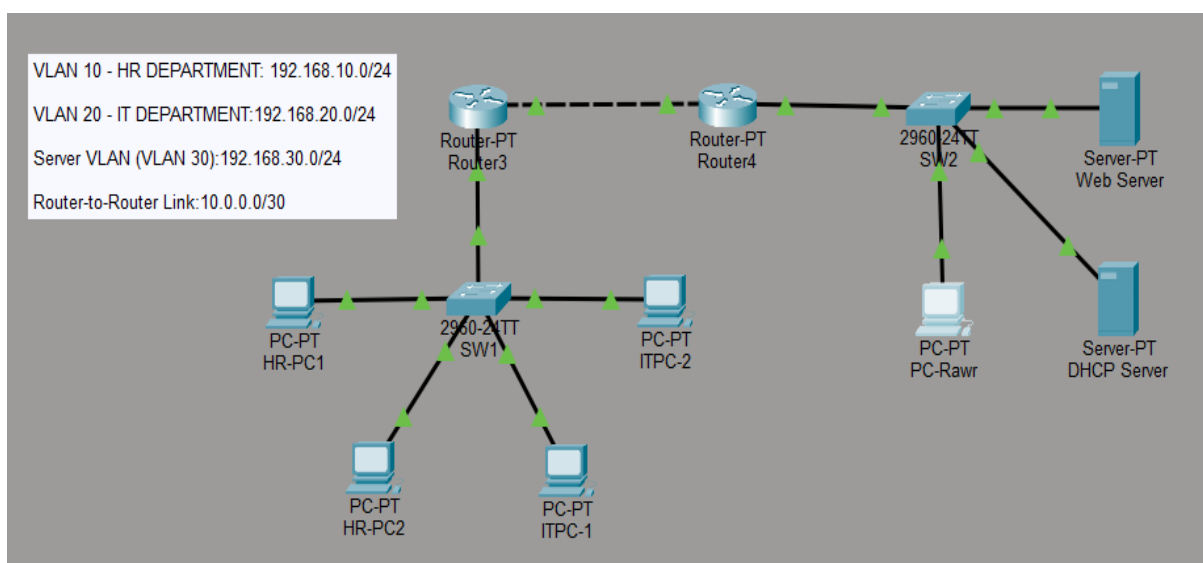


Network Design Overview

The network is built using two routers, two switches, servers, and several computers. Each device in the network has a specific job to make sure communication is smooth, organized, and secure. The routers are responsible for sending data between different networks, while the switches connect end devices such as computers and servers within the same local network. Router 3 is used to route traffic between different networks using dynamic routing with RIP. This allows the router to automatically learn routes and share routing information with other routers.

Router 4 is connected to Router 3 and is used to extend the network by connecting additional subnets. It also helps forward data packets to the correct destination. Two switches are used in the network to connect the computers and servers. These switches separate devices into different groups based on their department and help reduce unnecessary network traffic. By using switches, devices inside the same network can communicate faster and more efficiently. The network is divided into three main networks to improve organization and security.

These networks represent different departments and functions within the organization. The three networks are the HR Department Network, the IT Department Network, and the Server Network. Each network has its own IP address range to avoid conflicts and confusion. The Server Network contains important services such as a DHCP Server and a Web Server. The DHCP Server automatically assigns IP addresses to computers, which makes network setup easier and prevents IP address conflicts. The Web Server provides internal web services that can be accessed by allowed users in the network. One computer, PC – Rawr, is assigned the IP address **192.168.30.2**, which is used in the Access Control List configuration to control its access to the server.





Chosen Security Technique

The security technique used in this network is a **Standard Access Control List (ACL)**. An ACL is a group of rules configured on a router that controls network traffic. These rules decide which devices are allowed and which devices are not allowed to communicate with certain parts of the network. By using ACLs, network administrators can control access without changing the physical layout of the network. In this activity, the Standard ACL is configured on the router to manage traffic coming from specific devices.

Standard ACLs work by checking the **source IP address** of a computer before allowing or denying traffic. This makes them simple to configure and easy to understand, especially for basic network security setups. The ACL in this network is used to block one specific computer, **PC – Rawr**, from accessing other networks and protected resources. This computer has the IP address **192.168.30.2**, which is used in the ACL rule. When this computer tries to send traffic outside its network, the router checks the ACL and denies the connection.

All other computers that do not match the blocked IP address are still allowed to communicate normally. This shows how ACLs can block only one device without affecting the rest of the network. Through this configuration, the network becomes more secure while still allowing normal communication for authorized users.

Justification / Reasoning

The ACL is used to deny traffic coming from **PC – Rawr**, which has the IP address **192.168.30.2**, when it attempts to access other networks. When this computer sends data, the router checks the ACL rules first. If the source IP address matches the blocked address, the router automatically denies the traffic. This ensures that unauthorized traffic is blocked before it can reach protected networks. At the same time, the ACL allows all other computers and devices to communicate normally.

Only the specified IP address is restricted, while the rest of the network continues to function without problems. This shows how ACLs can be used to control access in a very specific way without affecting the entire network. Because the rule is applied on the router, it works for all connected networks and does not require changes on individual computers. This security method is useful in real-life situations where some computers should have limited access. For example, a guest computer, public



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workstation, or testing device should not be allowed to reach important department networks or servers.

By using an ACL, network administrators can easily restrict access while still keeping the network usable for authorized users. Overall, this technique helps protect sensitive information, prevents misuse, and improves the safety and control of the network.

Conclusion

In this activity, a secure multi-network design was successfully created using Cisco Packet Tracer. The network was built using routers, switches, servers, and multiple computers to simulate a real working environment. Routing was configured so that devices from different networks could communicate correctly. Servers were added to provide important services needed by users in the network. An Access Control List (ACL) was also implemented to control how devices are allowed to communicate.

By applying an ACL, the network became safer because unauthorized traffic was blocked before it could reach protected networks and servers. At the same time, other users were still able to communicate normally without interruption. This shows that security can be applied without affecting overall network performance. Through this activity, students were able to understand how routing, VLANs, and ACLs work together. Overall, the activity demonstrated that even simple security rules can greatly improve the protection, organization, and reliability of a computer network.

References

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